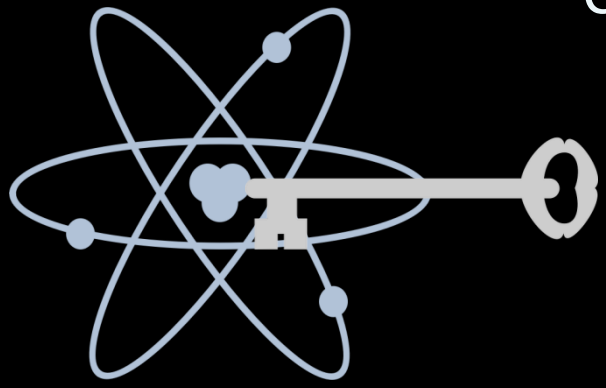
Quantum Homomorphic Encryption for Circuits of Low T-Gate Complexity

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Introduction

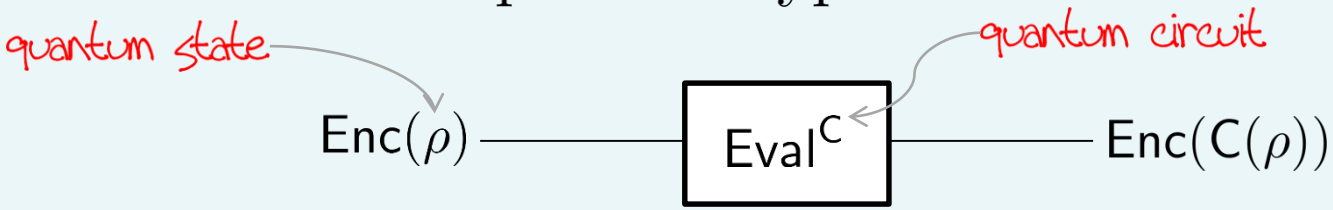
Classical Homomorphic Encryption

- Conceived in 1978 by Rivest, Adleman, and Dertouzos

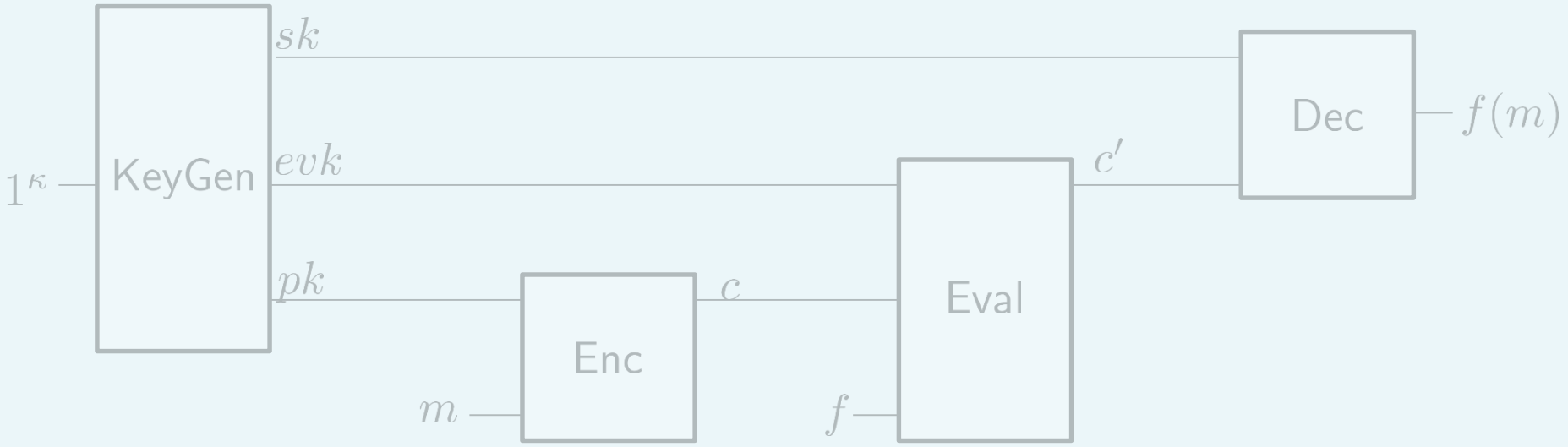


- *Fully* homomorphic encryption finally realized by Gentry in 2009

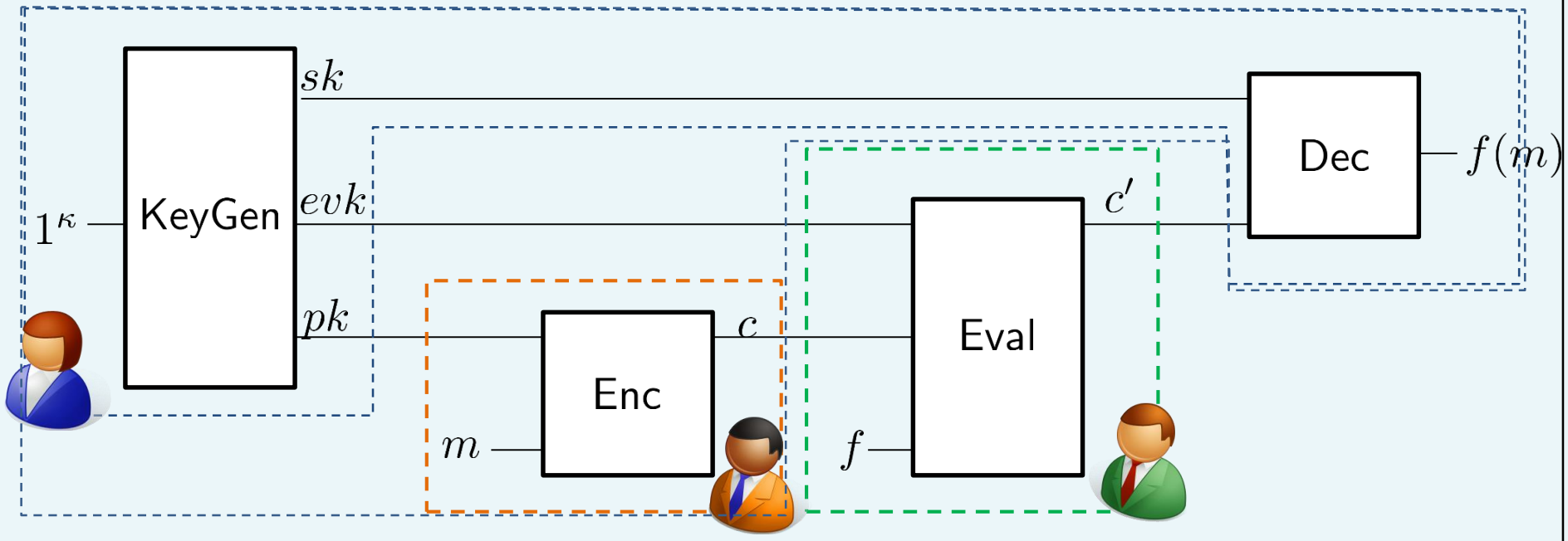
Quantum Homomorphic Encryption



Homomorphic Encryption



(Classical) Homomorphic Encryption



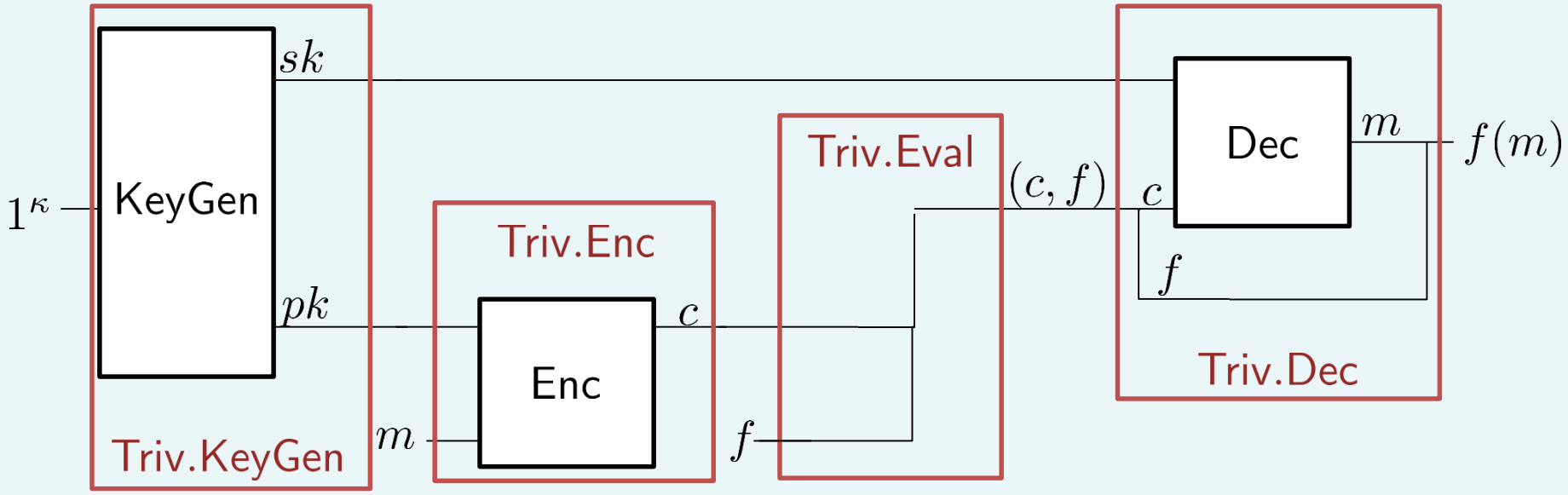
Properties:

$\mathcal{C}$-Homomorphic: for all $f \in \mathcal{C}$ $\mathsf{Dec}(sk, \mathsf{Eval}(evk, f, \mathsf{Enc}(pk, m))) = f(m)$

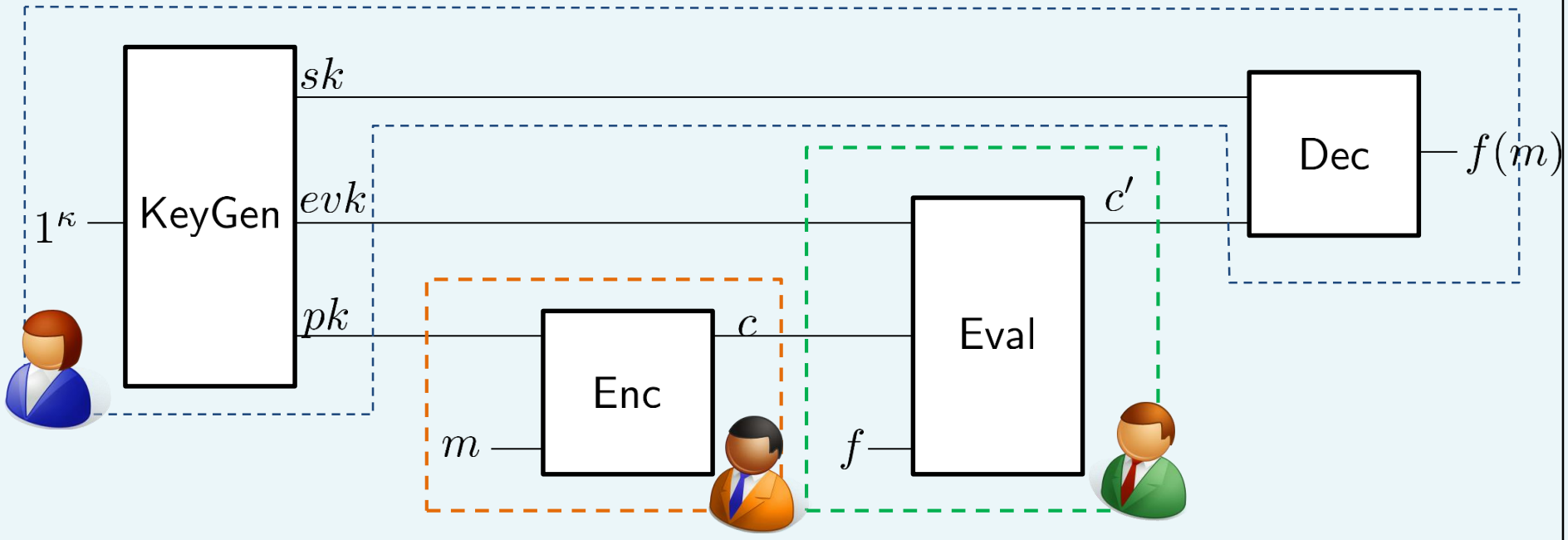
IND-CPA: Security: No poly-time adversary can distinguish $\mathsf{Enc}(0)$ from $\mathsf{Enc}(1)$

The Trivial Scheme

Let (KeyGen, Enc, Dec) be any PKE scheme.



(Classical) Homomorphic Encryption



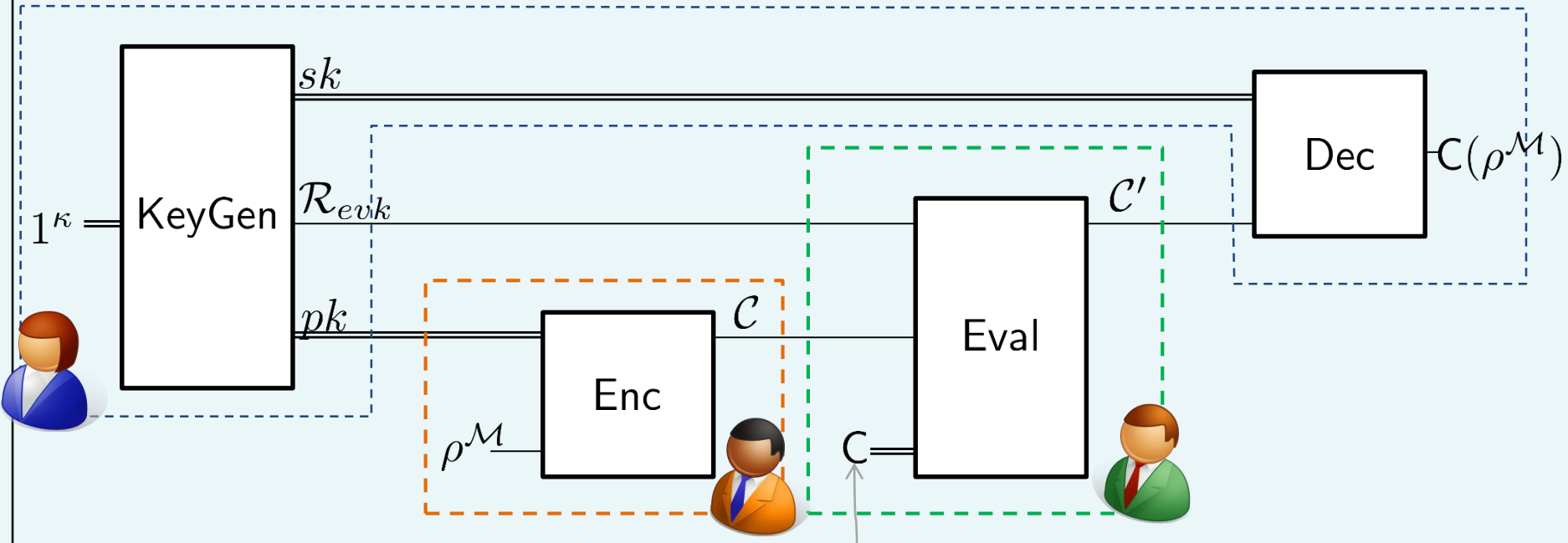
Properties:

$\mathcal{C}$-Homomorphic: for all $f \in \mathcal{C}$ $\mathsf{Dec}(sk, \mathsf{Eval}(evk, f, \mathsf{Enc}(pk, m))) = f(m)$

IND-CPA Secure: No poly-time adversary can distinguish $\mathsf{Enc}(0)$ from $\mathsf{Enc}(1)$

Compact: The complexity of decryption doesn't depend on f

Quantum Homomorphic Encryption



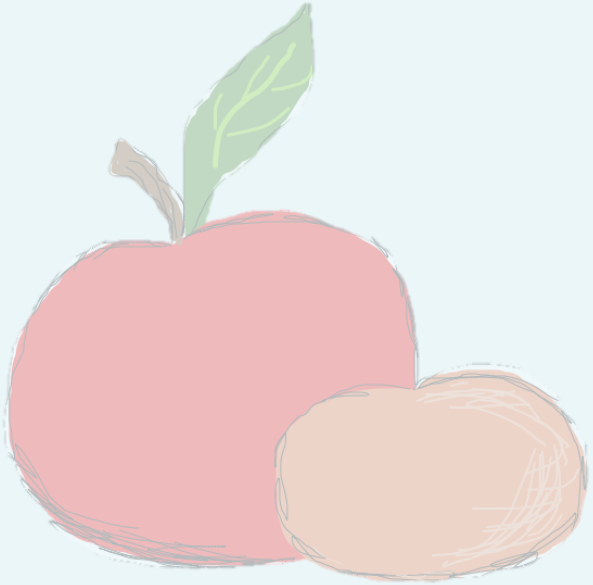
Properties:

$\mathcal{Q}$-Homomorphic: for all $\mathsf{C} \in \mathcal{Q}$ $\mathsf{Dec}_{sk}(\mathsf{Eval}^{\mathsf{C}}(\rho^{evk}, \mathsf{Enc}_{pk}(\rho^{\mathcal{M}}))) \approx \mathsf{C}(\rho^{\mathcal{M}})$

q-IND-CPA Secure: No poly-time adversary can distinguish $\mathsf{Enc}(\rho)$ from $\mathsf{Enc}(\$)$

Compact: complexity of Dec is independent of C

Our Results



Prior Related Work

- Secure delegated quantum computing
  - Childs 2005; Broadbent, Fitzsimons, Kashefi 2009; Aharonov, Ben-Or, Eban 2010; Broadbent 2015
  - All require interaction between encrypter and evaluator
- QFHE not possible with information theoretic security
  - Yu, Perez-Delgado, Fitzsimons 2014
- Quantum homomorphic encryption with information leakage (not semantic security)
  - Tan, Kettlewell, Ouyang, Chen, Fitzsimons 2014

Quantum vs. Classical Circuits

Classical Gates	Quantum Gates
Addition: $\oplus$	Cliffords: $\mathsf{H}, \mathsf{P},$ 
Multiplication: $\otimes$	Non-Clifford gate: T (*Any* non-Clifford gate would suffice)

Universal quantum gate set: $\{\mathsf{H}, \mathsf{P}, \mathsf{CNot}, \mathsf{T}\}$

Another universal quantum gate set: $\{\mathsf{H}, \mathsf{P}, \mathsf{CNot}, \mathsf{Tof}\}$

$\mathsf{P}\,|b\rangle = i^b\,|b\rangle \qquad \mathsf{T}\,|b\rangle = e^{i\pi b/4}\,|b\rangle$

Reversible Multiplication

Summary of Results

	Homomorphic?	Compact?	q-IND-CPA secure?
Not encrypting	quantum circuits	yes	no
Trivial Scheme	quantum circuits	complexity of Dec $\sim$ (# gates)	yes
Quantum One-time Pad	no	yes	yes
Our Results: Scheme 1: CL (public key)	Clifford circuits	yes	yes
Our Results: Scheme 2: AUX (symmetric key)	quantum circuits with const. T-depth	yes	yes
Our Results: Scheme 3: EPR (public key)	quantum circuits	complexity of Dec $\sim$ (# T-gates)^2	yes
Quantum Fully Homomorphic Encryption?	quantum circuits	yes	yes

Clifford Scheme



The Pauli Group

$$\mathsf{X}|0\rangle = |1\rangle \qquad \mathsf{Z}|+\rangle = |-\rangle$$
$$\mathsf{X}|1\rangle = |0\rangle \qquad \mathsf{Z}|-\rangle = |+\rangle$$

$$\vec{a} \in \{0,1\}^n \qquad \mathsf{X}^{\vec{a}} = \mathsf{X}^{a_1} \otimes \cdots \otimes \mathsf{X}^{a_n}$$

Pauli Group: $P_n \equiv \{\phi \mathsf{X}^{\vec{a}} \mathsf{Z}^{\vec{b}} : \vec{a}, \vec{b} \in \{0,1\}^n, \phi \in \{\pm i, \pm 1\}\}$

Quantum One-time Pad on n-qubit state:
Choose random $\vec{a}, \vec{b} \in \{0,1\}^n$, and apply $\mathsf{X}^{\vec{a}} \mathsf{Z}^{\vec{b}}$.

[AMTW00]

The Clifford Group

Clifford Group: Normalizer of the Pauli Group

$\forall$ Cliffords C, Paulis $\mathsf{X}^{\vec{a}}\mathsf{Z}^{\vec{b}}$

$\exists\ \vec{a}', \vec{b}'$ such that $\mathsf{C}\mathsf{X}^{\vec{a}}\mathsf{Z}^{\vec{b}} = \mathsf{X}^{\vec{a}'}\mathsf{Z}^{\vec{b}'}\mathsf{C}$ $\qquad (\vec{a}', \vec{b}') = f_{\mathsf{C}}(\vec{a}, \vec{b})$

linear function



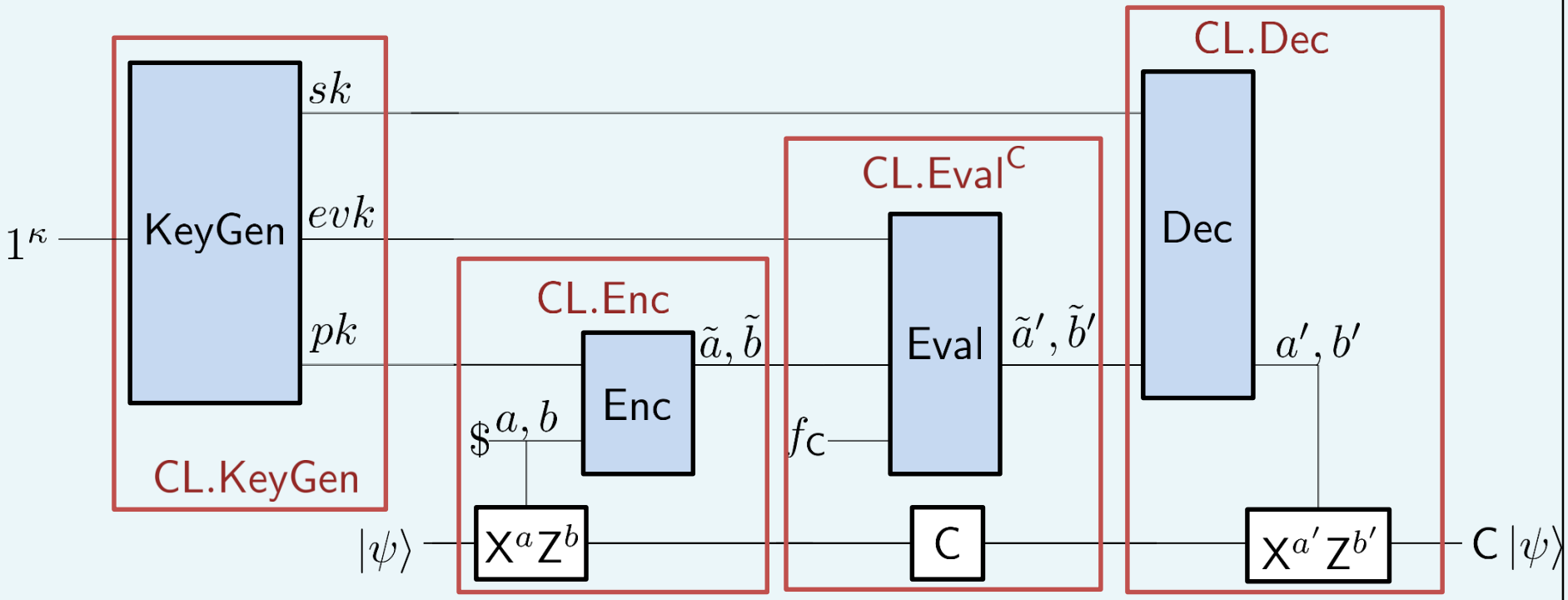
Example: $\mathsf{P}\mathsf{X}^{a}\mathsf{Z}^{b} = \mathsf{X}^{a}\mathsf{Z}^{a\oplus b}\mathsf{P}$ $\qquad f_{\mathsf{P}}(a, b) = (a, a \oplus b)$

Scheme 1: QHE for Cliffords (CL)

1) Quantum One-time pad: $|\psi\rangle \mapsto \mathsf{X}^a\mathsf{Z}^b\,|\psi\rangle$

2) For any Clifford C: $\mathsf{C}\mathsf{X}^a\mathsf{Z}^b\,|\psi\rangle = \mathsf{X}^{a'}\mathsf{Z}^{b'}\mathsf{C}\,|\psi\rangle$ $\qquad (a', b') = f_{\mathsf{C}}(a, b)$

3) Let (KeyGen, Enc, Eval, Dec) be a classical FHE scheme.



Summary of Results

	Homomorphic?	Compact?	q-IND-CPA secure?
Not encrypting	quantum circuits	yes	no
Trivial Scheme	quantum circuits	complexity of Dec $\sim$ (# gates)	yes
Quantum One-time Pad	no	yes	yes
Scheme 1: CL	Clifford circuits	yes	yes
Scheme 2: AUX	quantum circuits with const. T-depth	yes	yes
Scheme 3: EPR	quantum circuits	complexity of Dec $\sim$ (# T-gates)^2	yes
Quantum Fully Homomorphic Encryption	quantum circuits	yes	yes

Our Results

Summary of Results

	Homomorphic?	Compact?	q-IND-CPA secure?
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Trivial Scheme	quantum circuits	complexity of Dec	yes
Quantum One-time Pad	no		
Scheme 1: CL	Clifford circuits		
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Scheme 3: EPR	quantum circuits	complexity of Dec $\sim$ (# T-gates)^2	yes
Quantum Fully Homomorphic Encryption	quantum		

Our Results

Can use auxiliary states for T-gate evaluation.

For depth L, need n^{2^L} states!

Can use entanglement to delay part of T-gate evaluation until decryption.

Summary of Results

Our Results

	Homomorphic?	Compact?	q-IND-CPA secure?
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Quantum Fully Homomorphic Encryption?			yes

Thanks!